

I resume the PDC analysis by introducing several new issues. Firstly, Tot outputs were calibrated to be flat across all channels as follows. Here, Tot was selected as the highest one in the event with a gated He isotope region by HodP. Figures 205 and 206 show the Tot outputs distribution as a function of channels for each layer, where Tot was the maximum value in the plane, with and without calibration, respectively. The parameter file, including the calibration, is **20250901PDC_v01.xml**. Secondly, the Tot information was converted to the charge using the well-adapted equations;

$$Q = k_0 e^{+w/\tau},$$

where, $\tau=80$ nsec and 0.78125 nsec/ch were used. Third, CRM methods with pad integration of induced charges were implemented. Currently, the called 'effective gap' of 5.0 mm is used. After implementing these aspects, the calculated position and angle appear similar to those in the previous analysis. Thirdly, the clustering step was improved by including a sub-cluster.

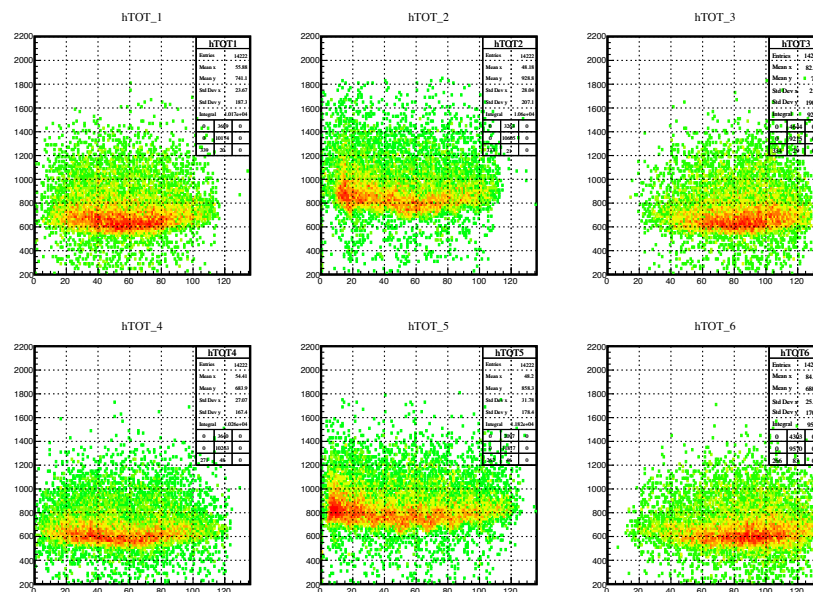


Figure 205, TOT distribution as a function of Ch of PDC on #310

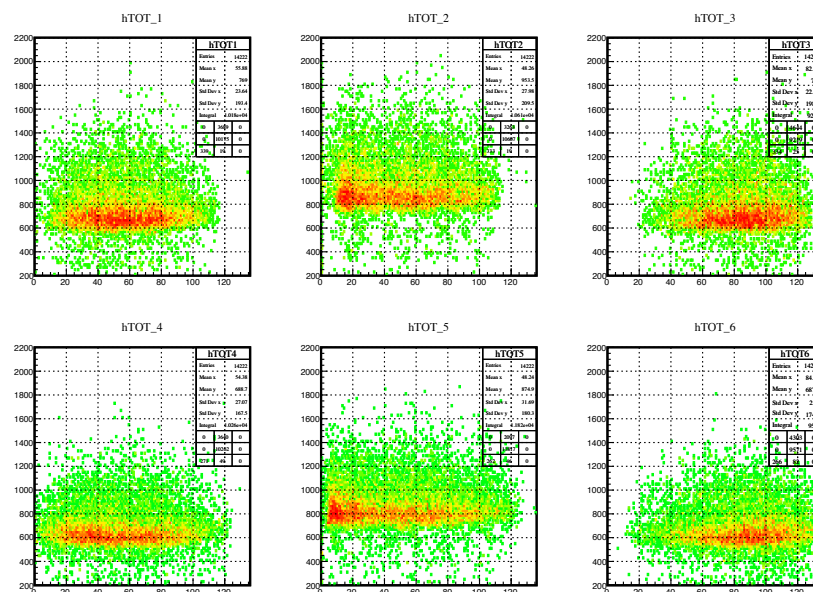


Figure 206, Same as Fig. 205 but after calibration #310.

The response function for CRM was adopted by SECHS, but it was integrated as;

$$q(X) = -\frac{Q}{2D} \int_{X-w/2}^{X+w/2} \frac{1}{\cosh \frac{\pi X'}{D}} dX' = -\frac{Q}{\pi} [\text{atan}(e^{\frac{\pi}{D}(X+w/2)}) - \text{atan}(e^{\frac{\pi}{D}(X-w/2)})].$$

The charge ratio $R = (q_i - q_{i+1}) / (q_i - q_{i-1})$ can give the position from the given value of experimental data.

One major change for 'charge' is how to add as the total Tot. The total Tot was calculated by

$$W^T = \sum W_i$$

becoming now by;

$$Q^T = \sum Q_i = \sum k_0 e^{W_i/\tau} \neq k_0 e^{W^T/\tau} = k_0 e^{(\sum W_i)/\tau}.$$

The following procedure was done in the PDC analysis as;

- 1, Collect Hit-Class as the usual step.
- 2, Filter the Hit events with TDC and ToT information, later converted to charge information.
- 3, Sorted the survival hits and clustered based on the channel information (neighbouring ones).
- 4, In the clusters, further categorizing a sub-cluster based on the charge information (up to three sub-clusters).
- 5, Deduce the position of each cluster with the selected method 1-4(3: SECHS, 4:CRM)
- 6, Calculate all the combinations in the 2-sets of U-V planes, then calculate chi-squared values, including X-planes, in addition to the U-V planes. Then, fill in Track-Class.
- 7, Sorted the Track-Class based on the chi-squared values.
- 8, Remove the track using the following used tracks with the worse chi-squared values. (Under study for improvement)

Figure 207 shows the correlation between the reconstructed position using cathode charges and the TDC of the perpendicular plane, which provides a rough position resolution using SECHS and CRM, respectively. The tracking using two U cathode planes was interpolated to the U anode planes, which were next to the V planes. TDC information of the V plane was mimicked by the timing of the **U anode wire**, whose spacing is **16/(2) mm** (on the other hand, **12 mm** (3 mm spacing x4 wires) step for **cathode wire**). The present analysis cannot yet provide a satisfactory resolution. Thus, we cannot conclude which method provides a better resolution at the moment.

Figure 208 shows the simple estimation based on Fig.207. The zigzag shape was corrected using the linear approximation with the reconstructed position. Currently, the resolution in the local region can be estimated to be approximately 0.5 mm in σ , after a reasonable correction.

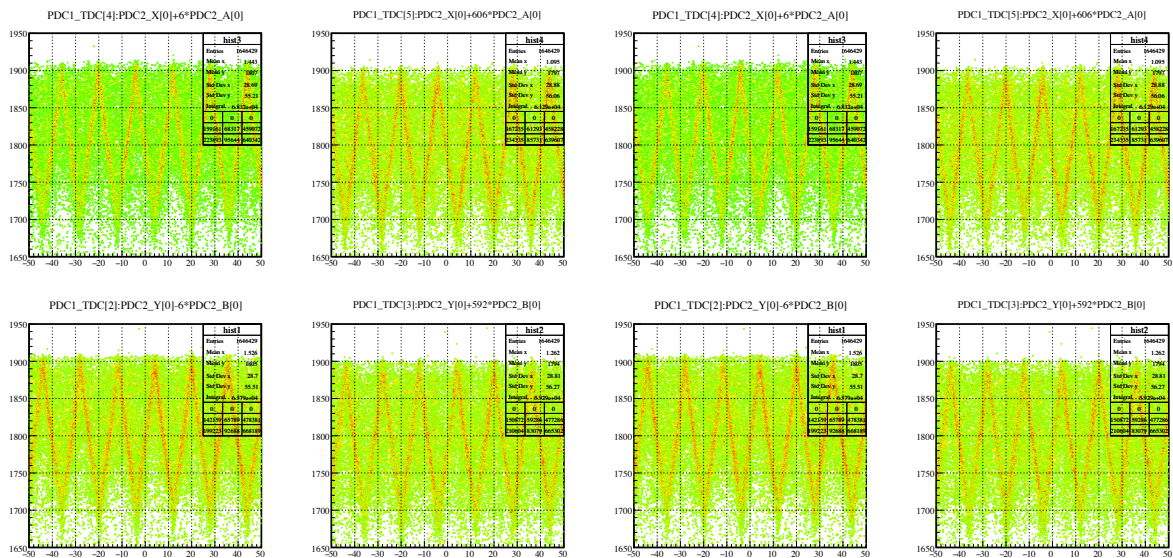


Figure 207, TDC correlation with reconstructed position by Cathode signal with SECHS(Left) and CRM(Right), same as Fig. 27.

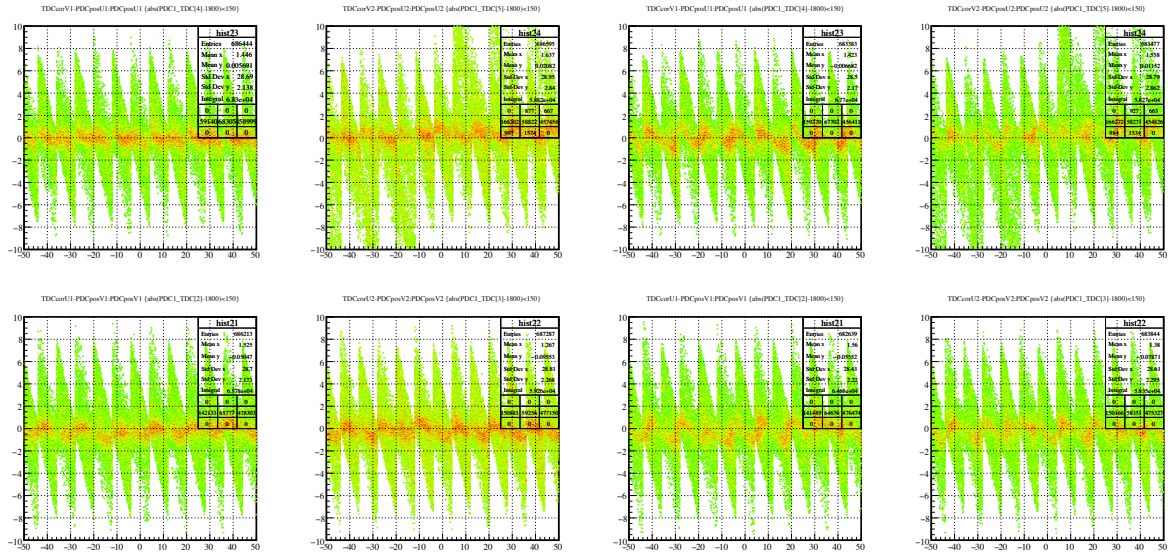


Figure 208, after subtracting the absolute position from Fig. 207.

>>How do we pick up the second track (or multiple tracks)?<<

In the present procedure, many fake (double-counting) events are included in the large ‘Chi-squared’ values. To avoid the phenomenon, firstly, I studied how to remove the event if the positions on the plane (U1, U2, V1, V2) were the same. Figure 209 shows the position correlation at HODP and PDC1 with gated HodpID = 30, under the conditions $\delta x = 1, 5, 10$, and 20. We can see the strong correlation in the position PDC1. If the position separation is only more than 1 mm, we can see the correlation even if it is 5mm. The right panel shows the result, which requires the same Tot for both tracks. It is also the unrealistic events, where some cluster is double-counting. If we apply the 10 mm, which is close to the step of the PDC cathode readout (12mm). No correlation remained. Thus, 10 mm and avoiding the same Tot events would be good criteria to reject unphysical combinations, which will be implemented in the code in the future.

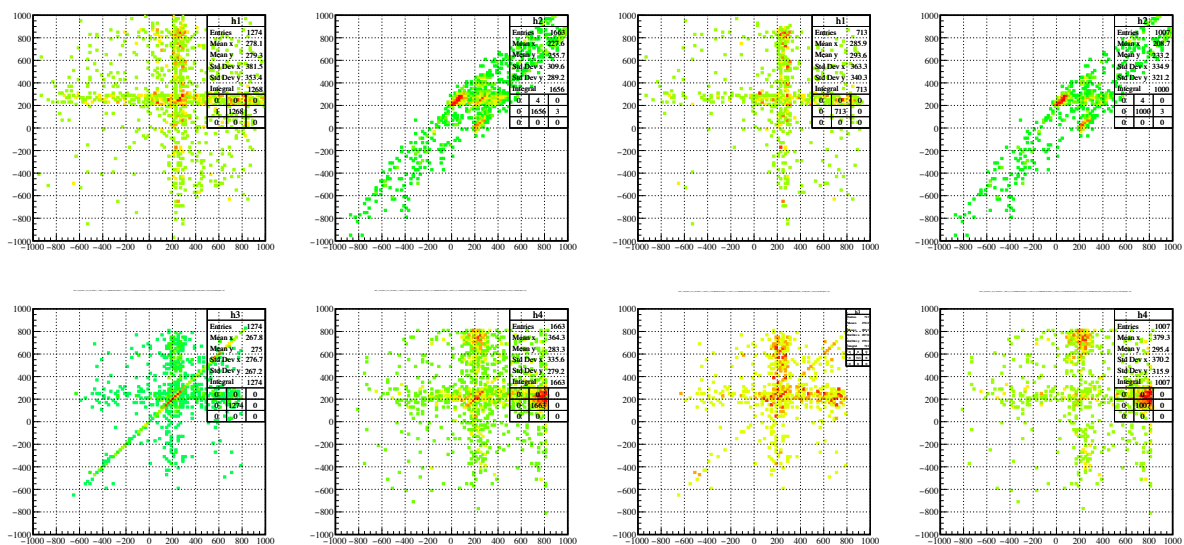


Figure 209, Position correlations on HodP(top) and PDC1(bottom) with different conditions on #309. The same Tot(Right) or not(Left) with coincidence of HodpID=30 and minimum position separation on the plane are 1mm(Left) and 5 mm(Right).

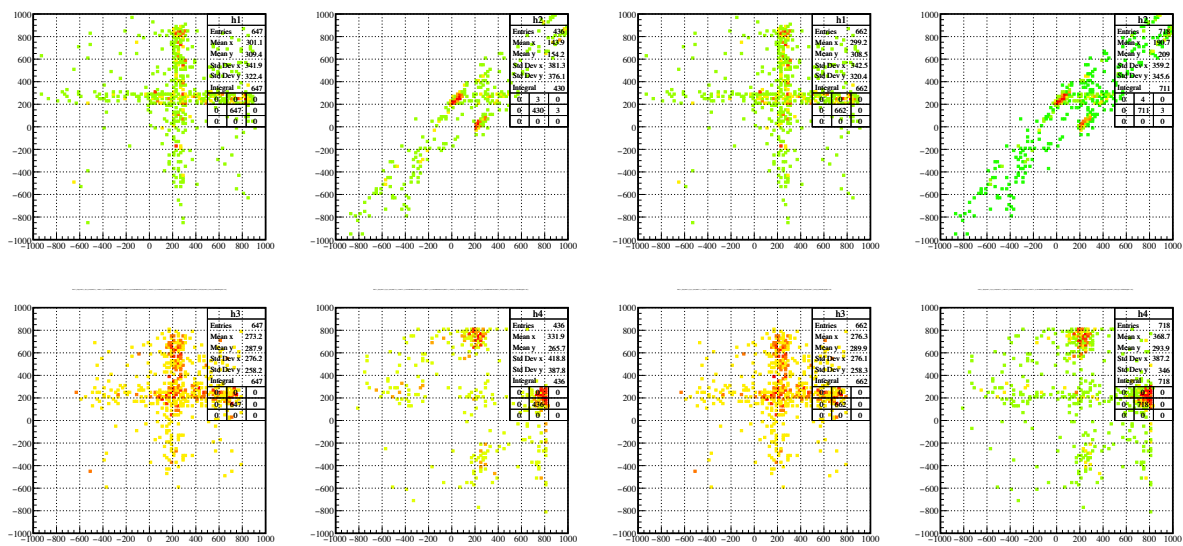


Figure 210, Same as Fig.209 but 10 mm(Left) and 20 mm(Right).

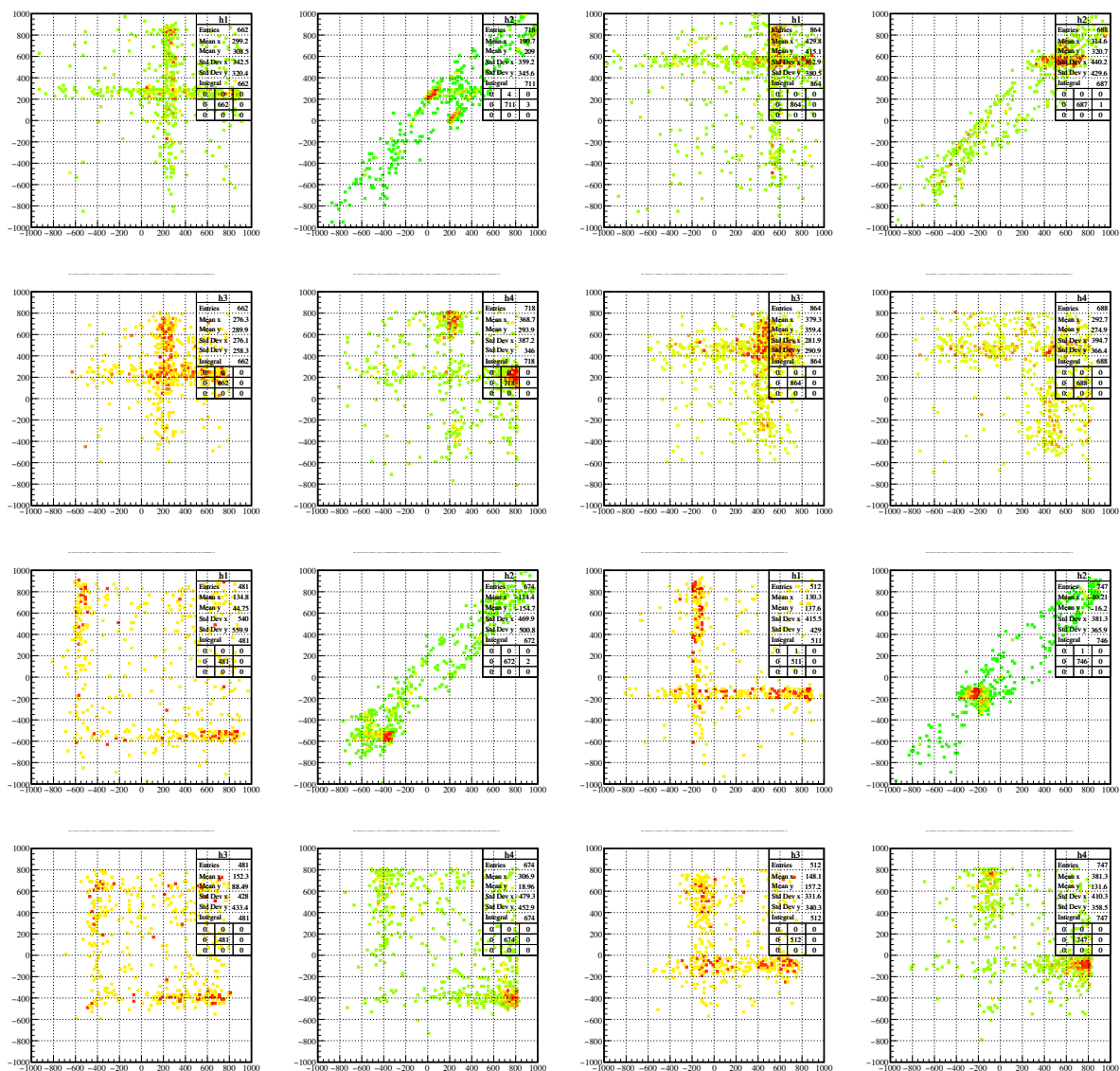


Figure 211, Same as Fig.209 and Fig.210. but different ID gate=27,30,34,38.

Next, we applied the different HodID conditions, successfully achieving a 10 mm separation. In all the cases, we can see a clear random correlation both on the HodP and the PDC1 in Fig.211. Figure 212 shows the same correlation plot without the HodplD gate. As we can see, the same Tot introduces the strange fake pattern on the HodP comes from the unnecessary event on high momentum side of 1st hit.

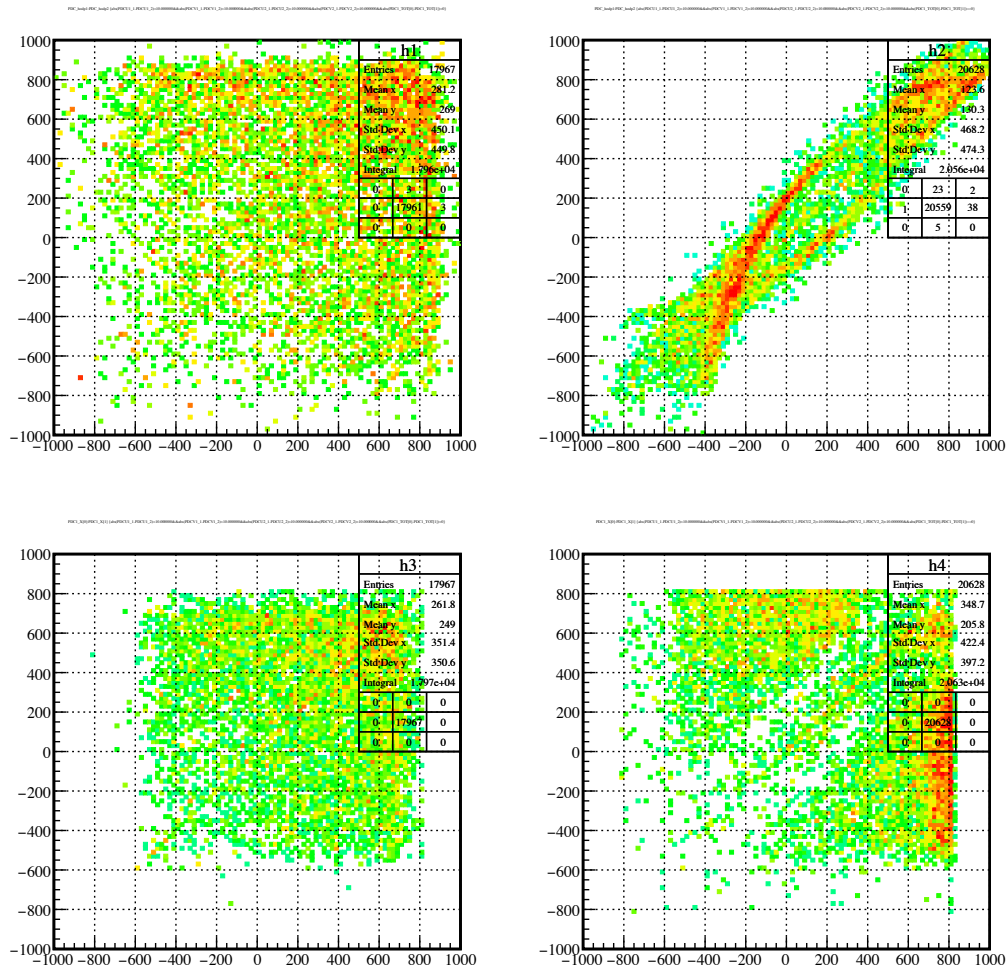


Figure 212, Same as above but no HodP conditions.

Figure 213 shows the correlation between the 1st and 2nd Chi distributions, as well as the distributions up to the 3rd one. No strange behavior was observed for correlation and distributions. Observed distribution more than 2-hit looks identical to the 1st hit without any conditions(Blue), after taking a reasonable scaling of 1/50. It means that the multiunit probability potentially has a few percent level of all the triggers. Figure 214 shows an example of the correlation between HodP and the PDC track. Currently, we still encounter many fake tracks.

I am still working to improve this type of analysis.

>>anaroot code files<<

Updated anaroot code can be found in anaroot04:~exp/exp2301_s053/anaroot/users/tera/git/samuraiana.git

It keeps updating through my private github. Updated (mainly PDC analysis) can be used on your requirement (2025/09/02).

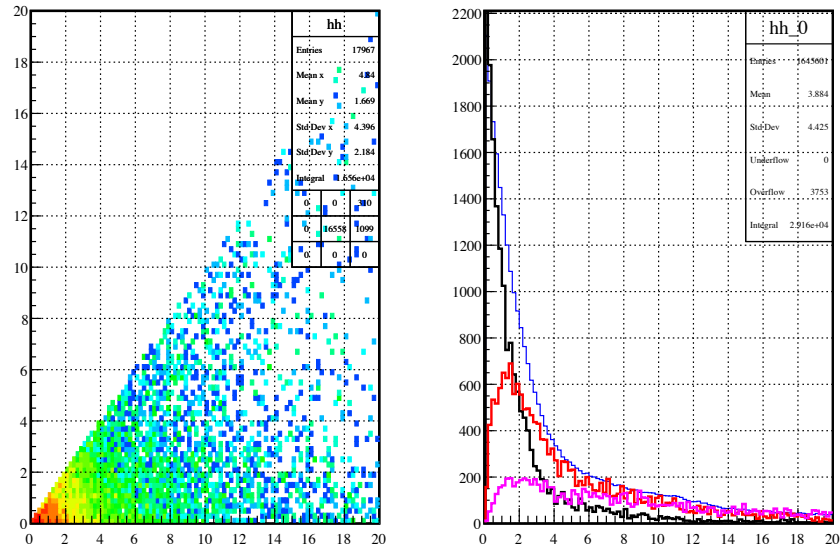
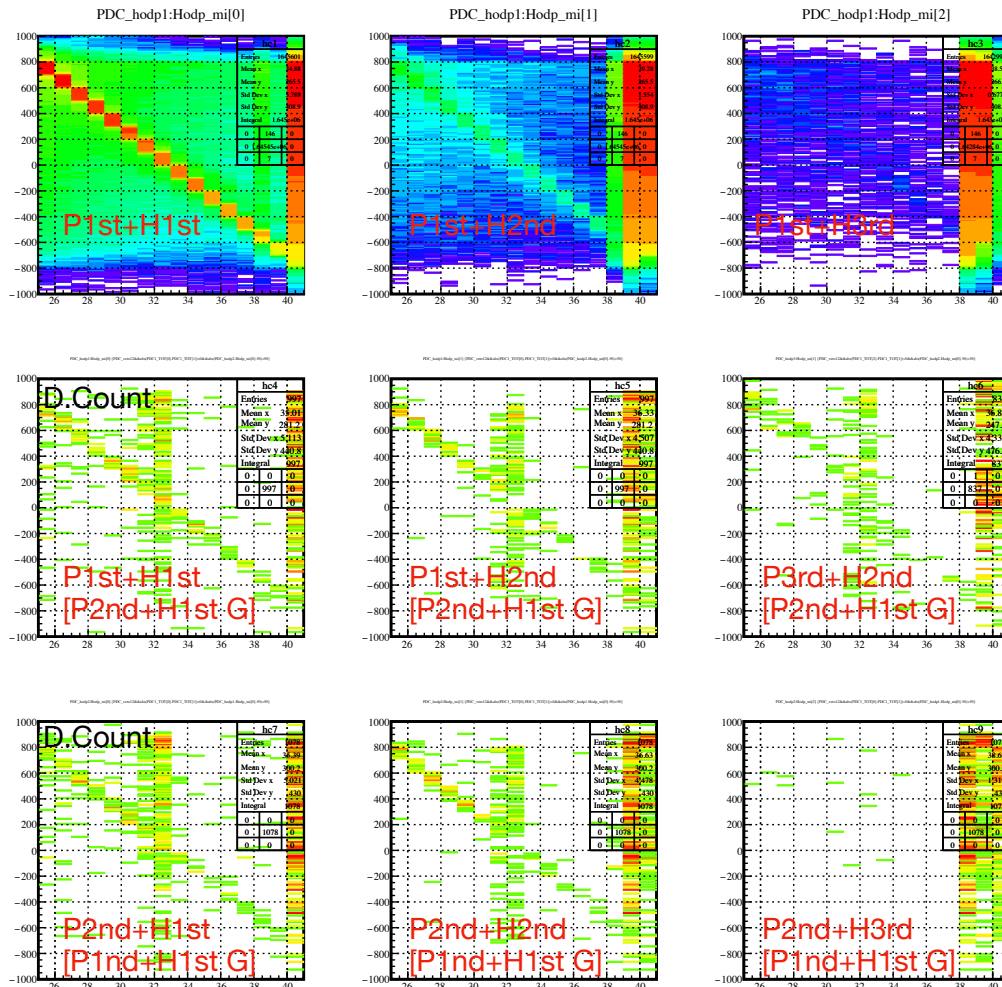


Figure 213, Observed Chi-square correlation between 1st and 2nd, and their distributions of



1st(Black), 2nd(Red), and 3rd(Purple), respectively.

Figure214, Example of the correlation between HodP and PDC on #309.